



COMSATS University Islamabad

Department of Computer Science

Course Description Form (CDF)

Course Information:

Course Code: AIC341

Course Title: **Introduction to Computer Vision**

Credit Hours: 3(2,1)

Lecture Hours/Week: 2

Lab Hours/Week: 3

Pre-Requisites: **None**

Course Objectives:

- To familiarize with both the theoretical and practical aspects of computing with images;
- To describe the foundation of image formation, measurement, and analysis;
- To gain exposure to object and scene recognition and categorization from images;
- To understand the geometric relationships between 2D images and the 3D world;
- To develop the practical skills necessary to build computer vision applications.

Course Content:

Overview; Basic Image Handling & Processing, Feature Detection, Feature Matching & Geometry, Motion & Tracking, Classical Object Detection, Deep Learning for CV, Emerging Topics

Unit wise Major Topics:

Unit	Topic	No. of teaching hours
1	Introduction to Computer Vision: Overview, Image Formation; Geometric Vision; CV Pipeline; and Applications.	3
2	Sampling and Multiscale Image Representations: Image Sampling & Aliasing; Down Sampling & Up Sampling Images; Image Features: Edges, Corners, & Blobs; Local Features & Key points; Hough Transform, and RANSAC	5
3	Image transformations and image warping; Stereo Vision; Depth from disparity; Epipolar Geometry; Stereo Calibration and Stereo Rectification	4
4	Model Fitting; Regularizer; Kernel Regression; Polynomial Curve fitting; Over fitting and under Fitting	4
5	Learning in Computer Vision: Supervised vs. Unsupervised learning in vision; Neural network; Regularization; Stochastic Gradient Descent; Optimizer; Back-propagation; Convolutional Neural Nets; CNN Architectures; U Net; Recurrent Neural Nets; Attention and Transformers.	7
6	Motion and Change Detection: Translational alignment Motion Estimation; 3D Motion & Its 2D Projection; Optical Flow Estimation, and Perception & Scene Understanding.	4
7	Generative Image Models and Representation Learning; Representation Learning; Perceptual Grouping; Generative Models; Generative Modeling Meets Representation Learning; and Conditional Generative Models	3
Total Contact Hours		30

Mapping of CLOs and GAs:

Sr.#	Unit #	Course Learning Outcomes	Blooms	GA
------	--------	--------------------------	--------	----

			Taxonomy Learning Level	
CLO's for Theory				
CLO-1	1	Explain fundamental concepts of Computer Vision and its applications.	<i>Understanding</i>	2
CLO-2	2, 3	Apply core techniques for image feature detection and matching	<i>Applying</i>	2,3
CLO-3	4, 5	Apply modern neural networks and probabilistic graphical models to represent and model visual patterns.	<i>Applying</i>	2,3
CLO-4	6,7	Analyze generative models and motion estimation techniques used in computer vision to interpret and model complex visual data.	<i>Analyzing</i>	2,3
CLO's for Lab				
CLO-5	2-7	Implement core computer vision algorithms using appropriate tools and libraries.	<i>Applying</i>	2,5
CLO-6	1-7	Develop computer vision application in a team environment.	<i>Creating</i>	2-6

CLO Assessment Mechanism:

Assessment Tools	CLO-1	CLO-2	CLO-3	CLO-4	CLO-5	CLO-6
Quizzes	Quiz 1	Quiz 2	Quiz 3	Quiz 4	-	
Assignments		Assignment 1	Assignment 2&3 (CCP ¹)	Assignment 4 (CCP)	Lab Assignments	
Sessional 1 Exam	Sessional 1 Exam				Lab Sessional 1	
Sessional 2 Exam		Sessional 2 Exam		-	Lab Sessional 2	
Final Term Exam	Final Terms Exam					
Final Term Exam	-	-	-	-	-	Lab Project/ Lab Final Term Exam (CCP+CCA ²)

1. CCP : Complex Computing Problem

2: CCA: Complex Computing Activity

Note:

1. Instructor should design theory assignments labeled as Complex Computing Problems (CCP) in such a way that each problem clearly exhibits 2–3 CCP characteristics.
2. Instructor should design theory assignments labeled as Complex Computing Activity (CCA) in such a way that each problem clearly exhibits 2–3 CCA characteristics.

Text and Reference Books:

Textbook:

1. Foundations of Computer Vision ; Antonio Torralba, Phillip Isola and William T. Freeman, MIT

Press, 2024

2. Computer Vision: Algorithms and Applications; Richard Szeliski, , Springer, 2022

Reference Book:

3. Computer Vision Using Deep Learning; Vaibhav Verdhan, , Apress, 2021